Assignment 3: Python Wrapper Test Case

In this assignment, an unsteady flat plate simulation was performed using the SU2 Python wrapper. The objective was to demonstrate how CFD simulations can be executed and controlled using Python, including the application of time-dependent boundary conditions.

Test Case Description:

The simulation considers incompressible flow over a flat plate, a classical benchmark problem for studying boundary layer behavior. The flow is characterized by low Mach number conditions ($M \approx 0.03$) and a Reynolds number of approximately 24,400.

The simulation is unsteady, with a time-dependent wall temperature applied on the flat plate. The wall temperature is defined as:

$$T_{wall} = 293 + 57 \sin(2\pi t)$$

This allows dynamic modification of boundary conditions during the simulation.

Configuration Details:

Key parameters used in the simulation:

Solver: RANS (Reynolds-Averaged Navier–Stokes)
Turbulence Model: SST (Shear Stress Transport)

- Time Stepping:
  - Time Step (Δt): 0.003
  - Total Time Iterations: 10
- Freestream Conditions:
  - Velocity: (1.0, 0.0, 0.0) m/s
  - Temperature: 293.15 K
- Boundary Conditions:
  - Isothermal Wall (plate)
  - Farfield boundary

The simulation was executed using a Python script that updates the wall temperature at each time step through the SU2 Python interface.

Implementation:

The Python wrapper (pysu2) was used to initialize the solver, apply boundary condition updates, and control the time integration loop. At each time step, the wall temperature was modified using a sinusoidal function, and the solver was advanced accordingly.

This demonstrates how SU2 can be integrated with Python to perform automated and customizable CFD simulations.

Results:

- Temperature contours at different time steps were obtained using ParaView. The results show minimal variation in temperature distribution due to the small simulation time and gradual change in the imposed boundary condition.

- These results confirm that the Python wrapper successfully applies time-dependent boundary conditions during the simulation.

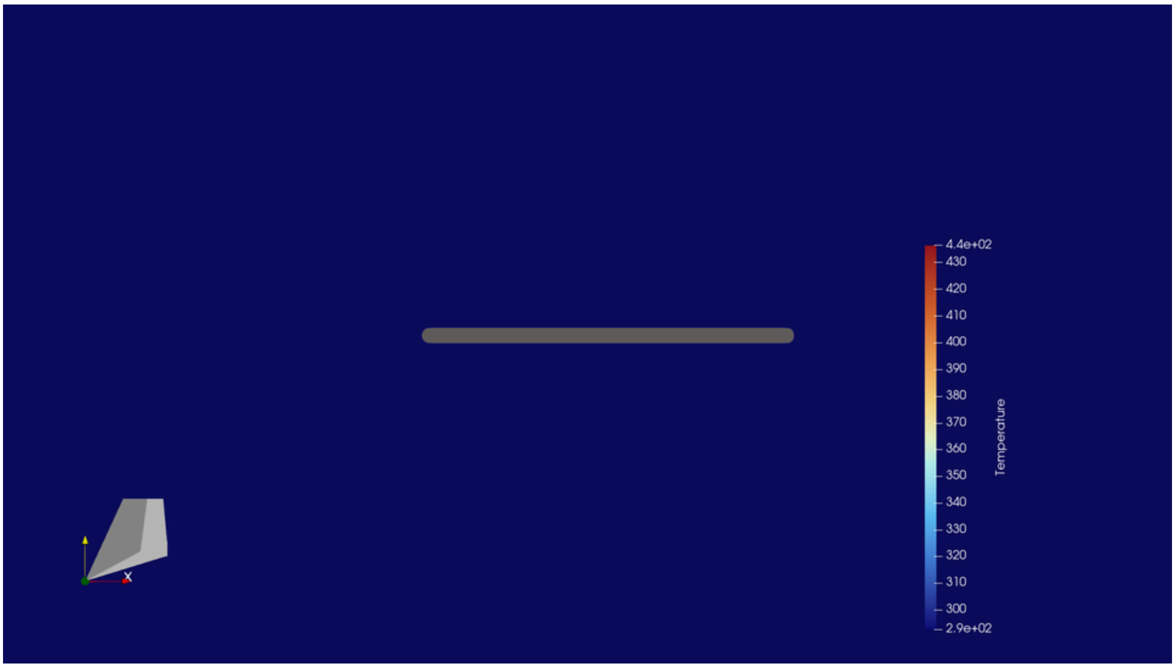
Conclusion:

The flat plate simulation was successfully executed using the SU2 Python wrapper. The study demonstrates the capability of applying dynamic boundary conditions and controlling CFD simulations programmatically.

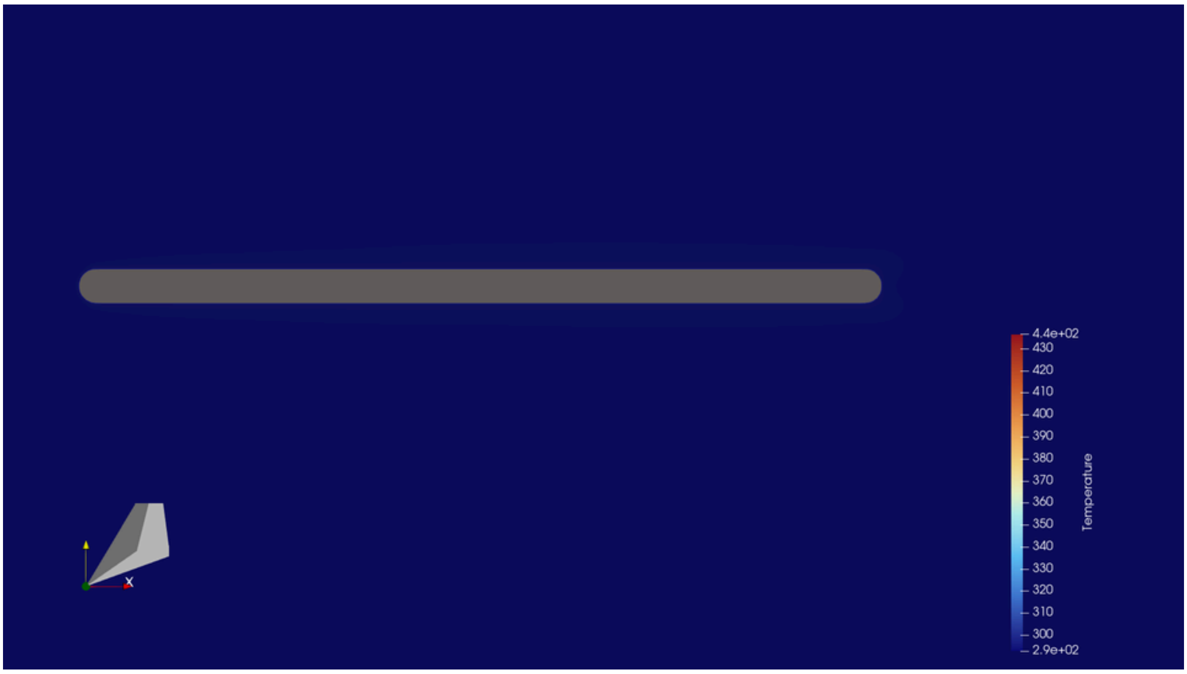
The results highlight the effectiveness of the Python interface for unsteady simulations and provide a foundation for more advanced CFD workflows involving automation and parameter control.

Outputs:

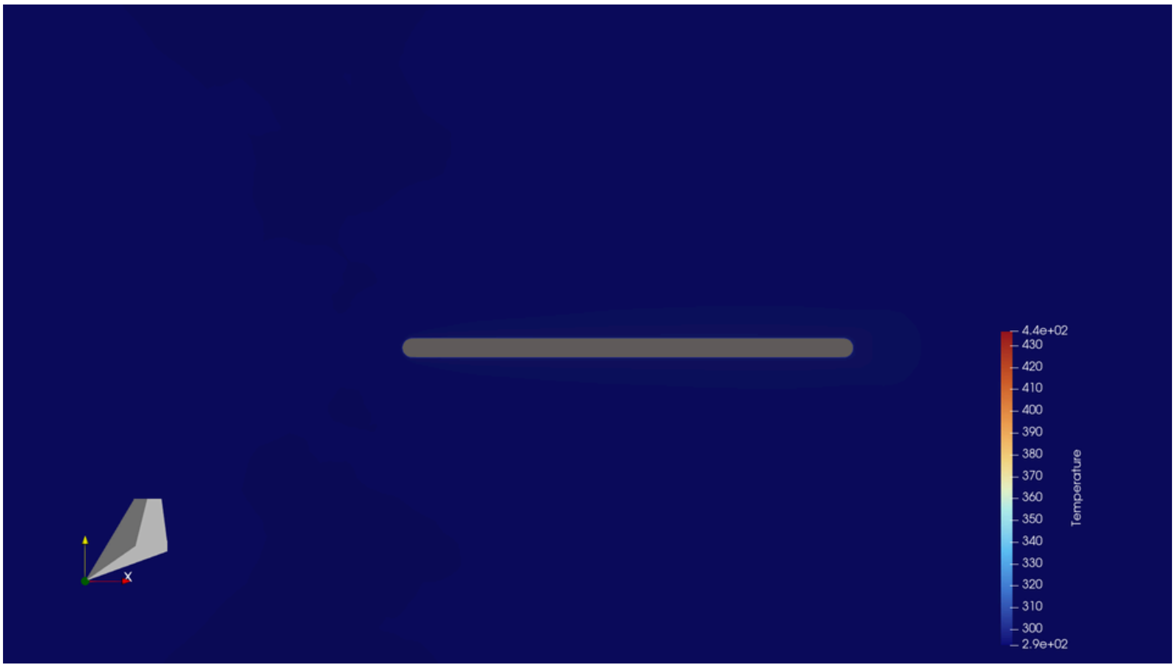
Temperature contour at iteration 0 (t ≈ 0 s)



Temperature contour at iteration 3 (t ≈ 0.009 s)



Temperature contour at iteration 6 (t ≈ 0.018 s)



Temperature contour at iteration 9 (t ≈ 0.027 s)